

```

1 clear;
2 clc;
3 close all;
4
5 %% PARAMETERS
6
7 num_vehicles = 10;      % Number of vehicles
8 num_slots = 20;        % Number of time slots/messages
9
10 % Latency parameters: Mean and Standard Deviation (ms)
11 %      Mean      Std
12 latency_params = [1.0  0.2; % DSRC
13                  2.0  0.5; % C-V2X
14                  0.5  0.1]; % 5G NR V2X
15
16 tech = {'DSRC', 'C-V2X', '5G NR V2X'};
17
18 latency_all = zeros(num_vehicles, ...
19                    num_slots, ...
20                    length(tech));
21
22 %% =====
23 % SIMULATE LATENCY
24 % =====
25
26 for t = 1:length(tech)
27
28     mean_delay = latency_params(t,1);
29
30     std_delay = latency_params(t,2);
31
32     for v = 1:num_vehicles
33
34         for s = 1:num_slots
35
36             % Propagation delay
37             prop_delay = mean_delay + ...

```

Run

Save

Output

Input

V2X LATENCY COMPARISON

Average latency per technology:

DSRC : 1.27 ms

C-V2X : 2.24 ms

5G NR V2X : 0.76 ms

[Execution complete with exit code 0]

